MODULE - 3

Equilibrium of Coplanar Concurrent Force System

A body is said to be in equilibrium when its static position or its motion with uniform velocity is unaltered by the system of forces acting on it.

Conditions of Equilibrium:

A body is said to be in equilibrium when it does not have any translatory or rotational moment. This means, when a body is in equilibrium under the action of coplanar non-concurrent forces, the following equations are satisfied:

① The algebraic sum of all the vertical forces should be equal to zero.

$$\Sigma V = 0$$

② The algebraic sum of all the horizontal forces should be equal to zero.

$$\Sigma H = 0$$

(5) The algebraic sum of the moments of the forces about any point along the plane will be equal to zero.

$$\Sigma M_A = 0$$

Equilibrant

Sometimes, the resultant of the force system is not equal zero. That means, the body is not in equilibrium. The force which is required to keep the body in equilibrium is known as 'Equilibrant'.

Types of forces acting on a body :-

Various forces acting on a body may be classified as:

① Applied forces ② Non-Applied forces

① Applied forces :

The forces applied externally on a body having its own point of contact on a body is called a Applied force.

② Non-Applied forces :

There are 2-types of non-applied forces.

- Self weight
- Reaction

Self weight :- Every body is subjected to a gravitational attraction and hence has got self weight. The self weight can be given by the expression $\boxed{W = mg}$.

Where, m = Mass of the body

g = Gravitational Attraction ($9.81\ m/s^2$)

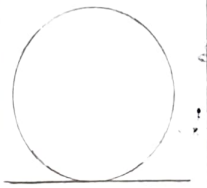
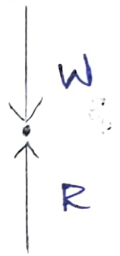
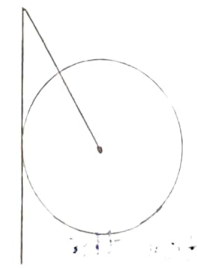
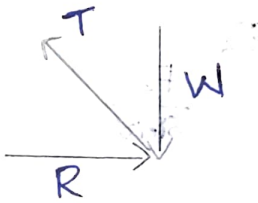
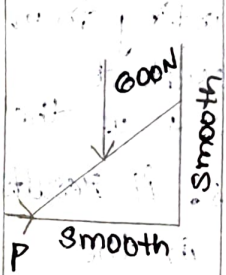
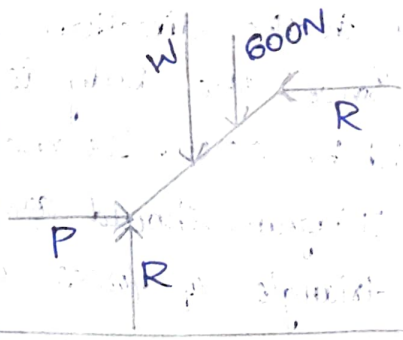
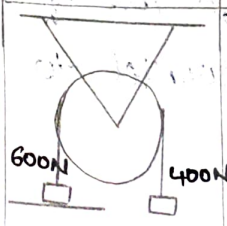
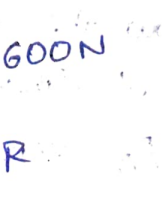
Reaction :- Reactions are the self adjusting forces developed by other bodies which are in contact with the body under consideration.

A/c to Newton's 3rd Law of motion,
the reactions are equal and opposite to actions.

Free Body Diagram:

Free Body Diagram of the body is a diagram in which the body is isolated from all its support points and the forces acting on it [including self weight, reactions and Applied forces]. The above forces are indicated excluding the body and the support.

The below table shows the FBD for the few cases:

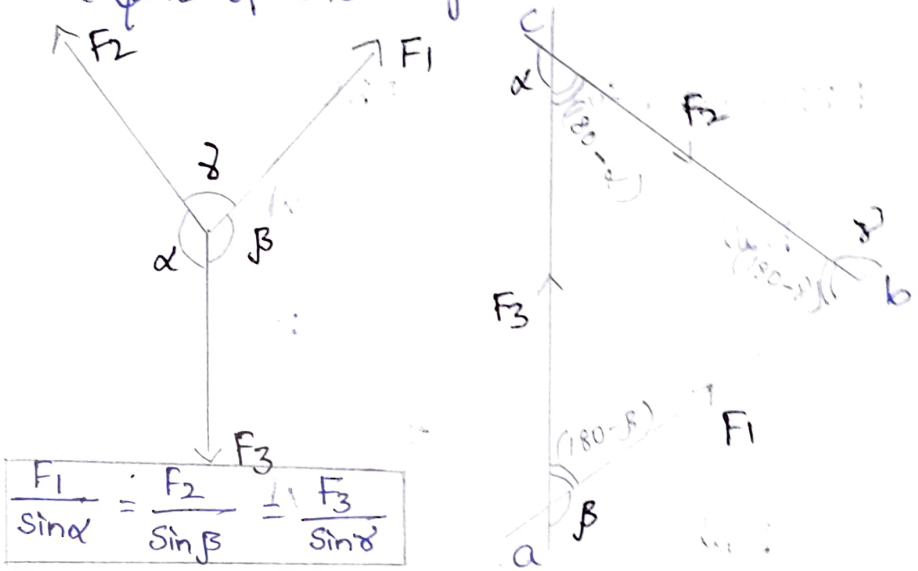
Reacting Bodies	FBD required for	FBD
	Ball	
	Ball	
	ladder	
	Block weighing 600N	

LAMI's Theorem:

If a body is in equilibrium under the action of coplanar concurrent forces it may be analysed using equations of equilibrium. However if a body is in equilibrium under the action of only three forces, LAMI's Theorem can be used conviniently.

Statement :-

"If a body is in equilibrium under the action of only three (3) forces, Each force is proportional to the sine of the angle between the other two forces."



$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

Proof :-

Draw the three forces F_1, F_2 & F_3 one after the other in the direction and magnitude starting from 'a'. Since the body is in equilibrium, Resultant should be zero. It means, the last point of the Force Diagram should coincide with 'a'. Thus, it results in a triangle of forces a, b & c as shown in the figure. Now the external angles at a, b, c are equal to β, γ & α since ab, bc & ca are Parallel to F_1, F_2 & F_3 respectively.

In the Δ^{le} of forces abc

$ab = F_1$

$bc = F_2$

$ca = F_3$

Applying the Sine Rule for the Δ^{le} abc

$$\frac{ab}{\sin(180-\alpha)} = \frac{bc}{\sin(180-\beta)} = \frac{ca}{\sin(180-\gamma)}$$

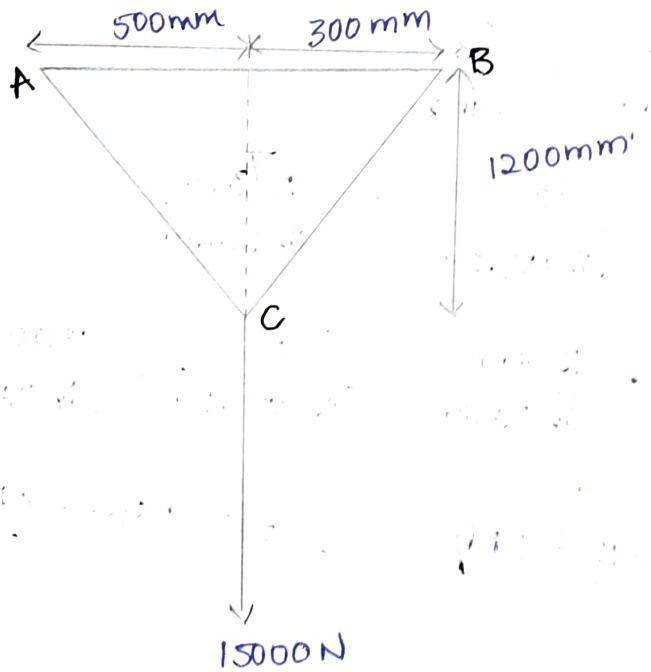
Now as per trigonometric law,

$$\sin(180-\theta) = \sin\theta$$

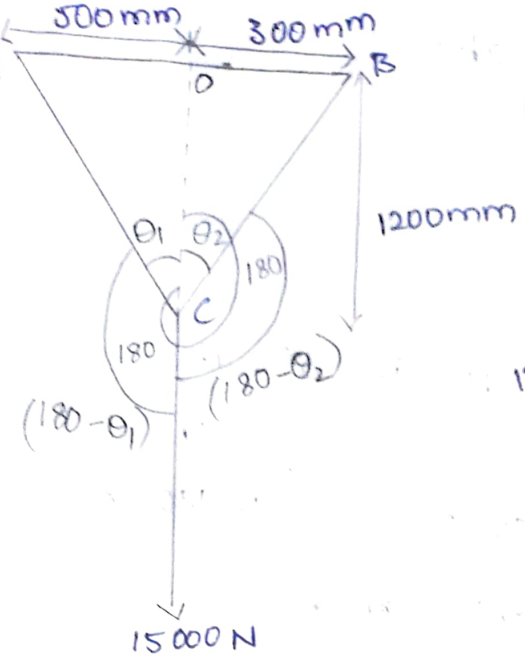
$$\therefore \boxed{\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}}$$

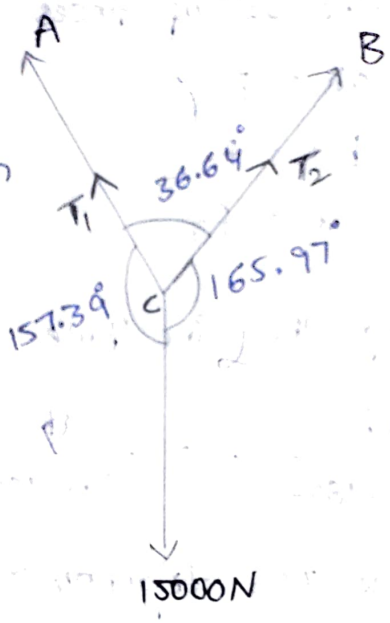
Problems

① Two cables are tied together at a point and loaded as shown in the figure. Determine the tensions in the cables AC & BC.



Sol





From Δ^{le} AOC

$$\tan\theta_1 = \frac{500}{1200}$$

$$\boxed{\theta_1 = 22.61^\circ}$$

From Δ^{le} COB

$$\tan\theta_2 = \frac{300}{1200}$$

$$\boxed{\theta_2 = 14.03^\circ}$$

Applying Lami's Theorem,

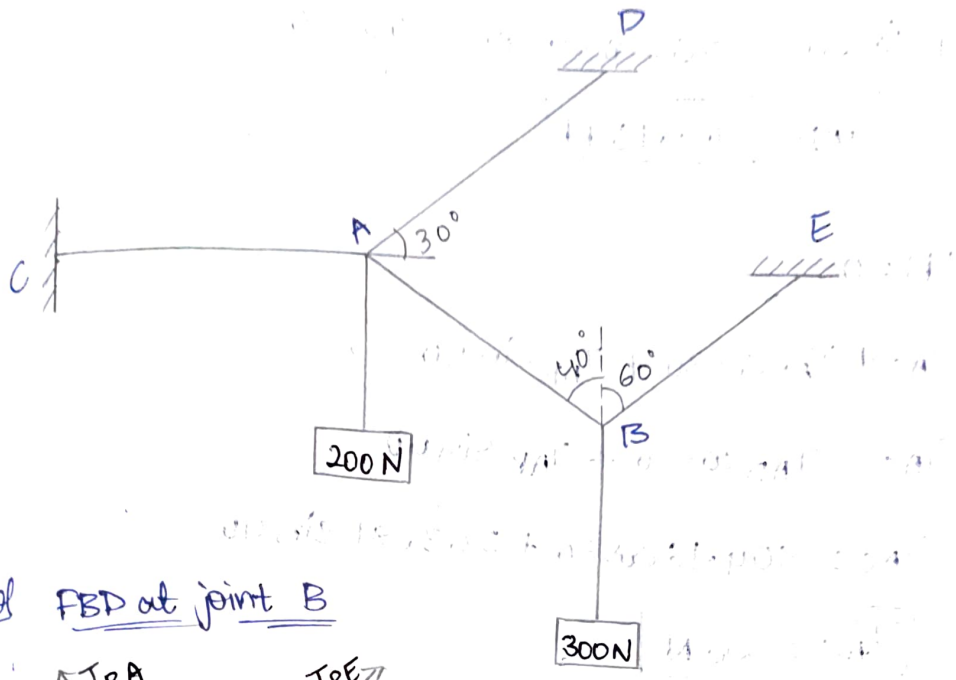
$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

$$\frac{T_1}{\sin 165.97} = \frac{T_2}{\sin 157.39} = \frac{15000}{\sin 36.64}$$

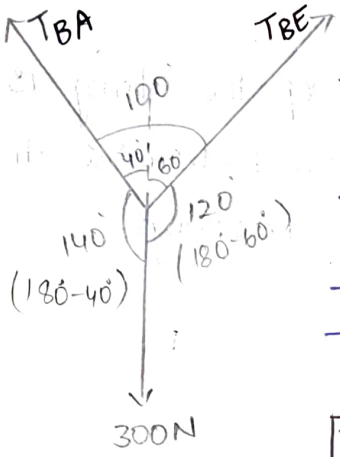
$$\frac{T_1}{\sin 165.97} = \frac{15000}{\sin 36.64} \qquad \frac{T_2}{\sin 157.39} = \frac{15000}{\sin 36.64}$$

$$\boxed{T_1 = 6098.39\ N} \qquad \boxed{T_2 = 9663.18\ N}$$

② Determine the forces in the wires shown in figure which are under equilibrium.



Sol FBD at joint B



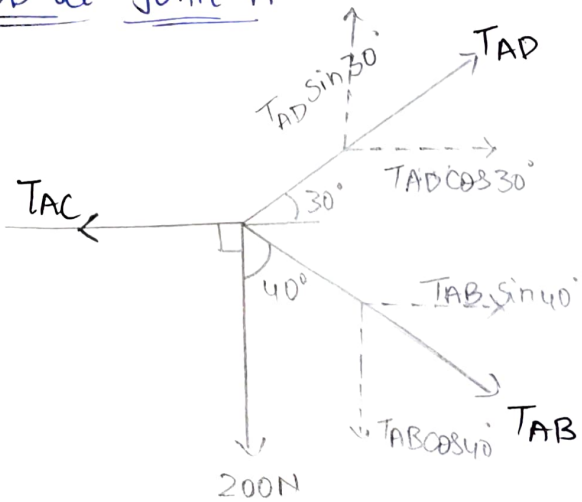
Applying Lami's Theorem,

$$\frac{T_{BA}}{\sin 120} = \frac{T_{BE}}{\sin 140} = \frac{300}{\sin 100}$$

$$\frac{T_{BA}}{\sin 120} = \frac{300}{\sin 100} \qquad \frac{T_{BE}}{\sin 140} = \frac{300}{\sin 100}$$

$T_{BA} = 263.81\,N$ $T_{BE} = 195.81\,N$

FBD at Jonit A



$\Sigma V = 0$

$T_{AD} \sin 30 - T_{AB} \cos 40 - 200 = 0$

$T_{AD} \sin 30 - 263.8 \cos 40 - 200 = 0$

$\boxed{T_{AD} = 804.16\ N}$

$\Sigma H = 0$

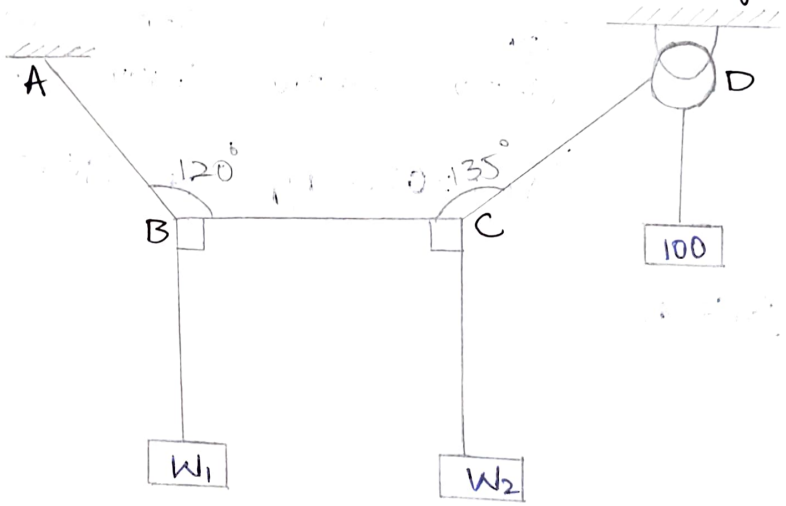
$-T_{AC} + T_{AD} \cos 30 + T_{AB} \sin 40 = 0$

$T_{AC} = T_{AD} \cos 30 + T_{AB} \sin 40$

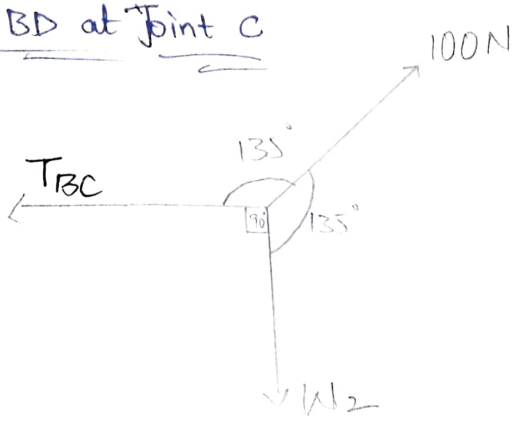
$T_{AC} = 804.16 \cos 30 + 263.81 \sin 40$

$\boxed{T_{AC} = 866\ N}$

③ In the figure, the portion BC of the string is horizontal and pulley is frictionless. Determine the tensions in the different parts of the strings.



Sol FBD at joint C



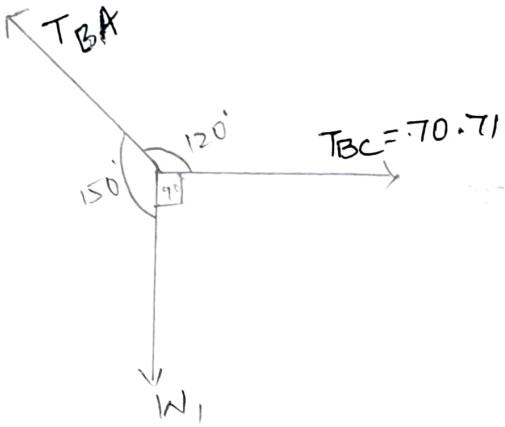
Applying Lami's Theorem

$$\frac{T_{BC}}{\sin 135} = \frac{W_2}{\sin 135} = \frac{100}{\sin 90}$$

$$\frac{T_{BC}}{\sin 135} = \frac{100}{\sin 90} \qquad \frac{W_2}{\sin 135} = \frac{100}{\sin 90}$$

$$\boxed{T_{BC} = 70.71\ N} \qquad \boxed{W_2 = 70.71\ N}$$

FBD at Joint B



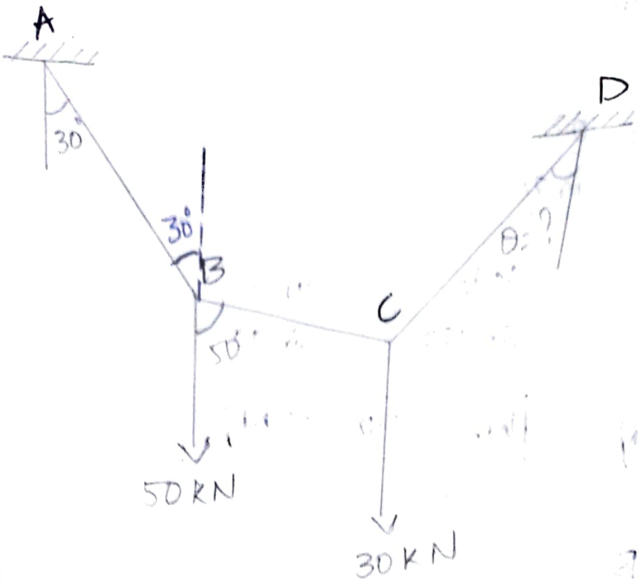
Applying Lami's Theorem

$$\frac{T_{BA}}{\sin 90} = \frac{T_{BC}}{\sin 150} = \frac{W_1}{\sin 120}$$

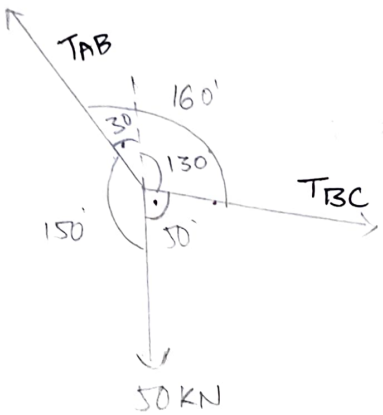
$$\frac{T_{BA}}{\sin 90} = \frac{70.71}{\sin 150} \qquad \frac{W_1}{\sin 120} = \frac{70.71}{\sin 150}$$

$$\boxed{T_{BA} = 141.4\ N} \qquad \boxed{W_1 = 122.47\ N}$$

(4) Determine the angle 'θ' for the system of strings ABCD in equilibrium as shown in the figure.



Sol FBD at Joint B



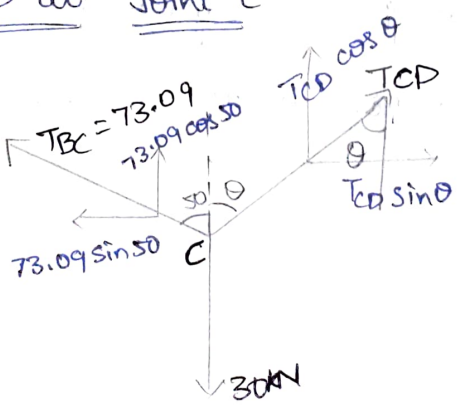
Applying Lami's theorem

$$\frac{T_{BC}}{\sin 150^\circ} = \frac{T_{AB}}{\sin 50} = \frac{50}{\sin 160^\circ}$$

$$\frac{T_{BC}}{\sin 150^\circ} = \frac{50}{\sin 160^\circ} \qquad \frac{T_{AB}}{\sin 50} = \frac{50}{\sin 160^\circ}$$

$$\boxed{T_{BC} = 73.09\ N} \qquad \boxed{T_{AB} = 111.98\ N}$$

FBD at Joint C



Applying equation of equilibrium

$\Sigma V = 0$

$80 + 73.09 \cos 50 + T_{CD} \cos\theta = 0$

$T_{CD} \cos\theta = -16.98 \longrightarrow (1)$

$\Sigma H = 0$

$73.09 \sin 50 + T_{CD} \sin\theta = 0$

$T_{CD} \sin\theta = 55.99 \longrightarrow (2)$

Now, $\frac{Eq^n (2)}{Eq^n (1)}$

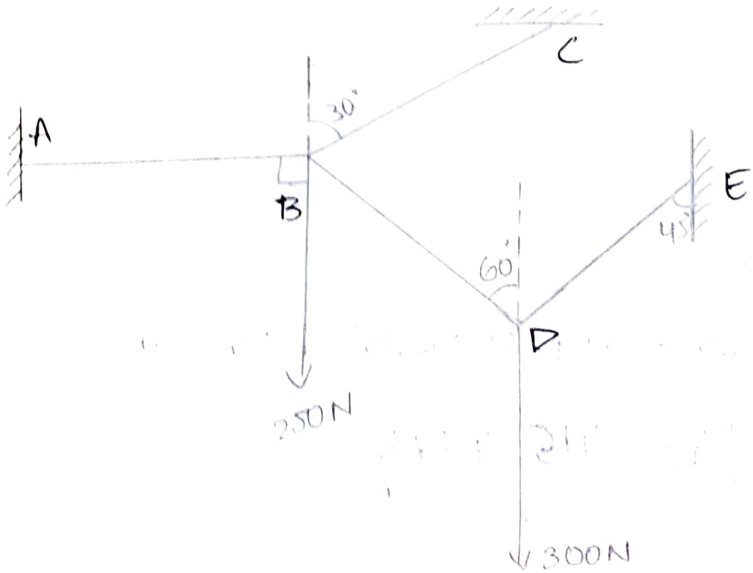
$$\frac{T_{CD} \sin\theta}{T_{CD} \cos\theta} = \frac{55.99}{-16.98}$$

$$\tan\theta = \frac{55.99}{-16.98}$$

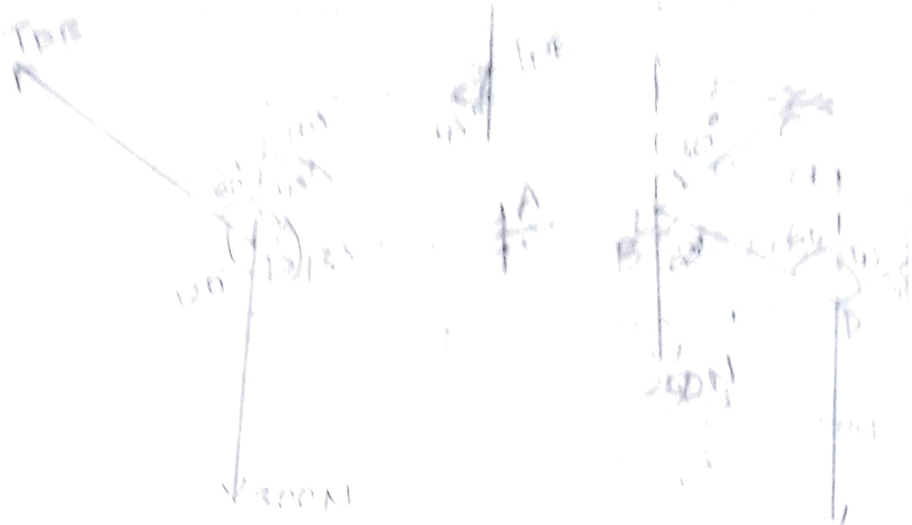
$$\theta = \tan^{-1}\left[\frac{55.99}{-16.98}\right]$$

$$\boxed{\theta = -73.12}$$

(5) A system of connected cables is shown in the figure supporting two vertical forces 250 N & 300 N at the point B & D as shown in the figure. Determine the forces in the various segments of the cable.



(b) FBD at Joint D



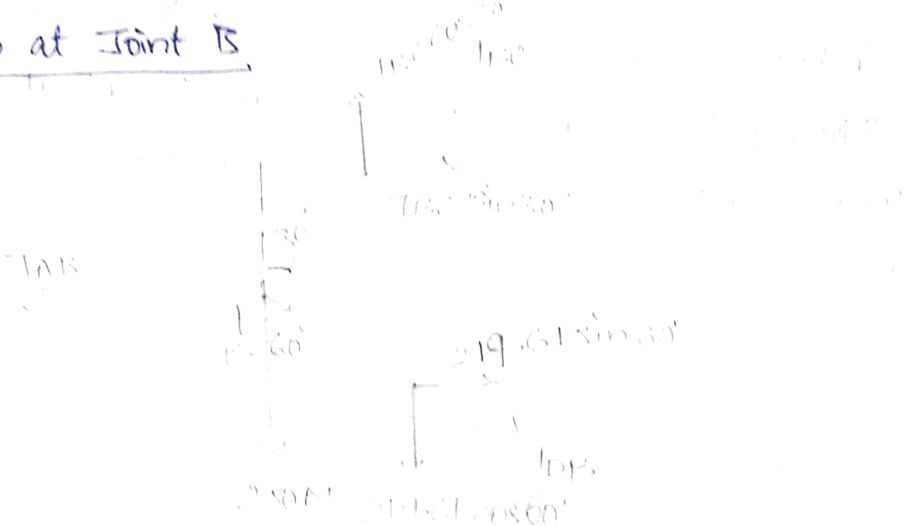
Apply Lami's Theorem

$$\frac{T_{DB}}{\sin 135^\circ} = \frac{T_{DE}}{\sin 120^\circ} = \frac{300}{\sin 105}$$

$$\frac{T_{DB}}{\sin 135^\circ} = \frac{300}{\sin 105} \qquad \frac{T_{DE}}{\sin 120^\circ} = \frac{300}{\sin 105}$$

$$\boxed{T_{DB} = 219.61\ N} \qquad \boxed{T_{DE} = 268.97\ N}$$

FBD at Joint B



$$\Sigma V = 0$$

$$T_{BC}\cos 30 - 219.61\cos 60^\circ - 250 = 0$$

$$\boxed{T_{BC} = 415.46\ N}$$

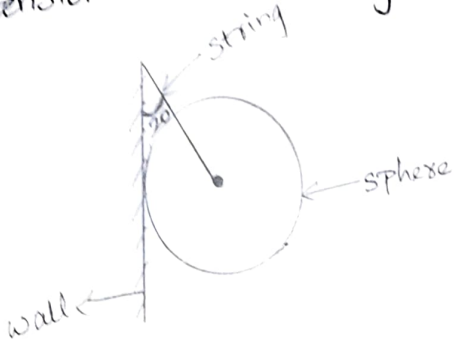
$\Sigma H = 0$

$T_{BC} \sin 30 + 219.61 \sin 60 - T_{AB} = 0$

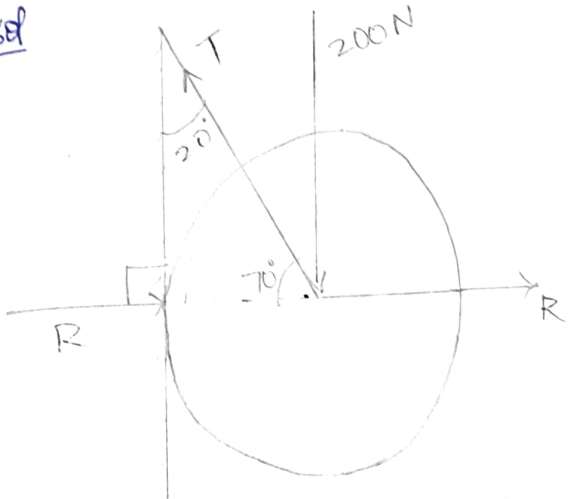
$415.46 \sin 30 + 219.61 \sin 60 = T_{AB}$

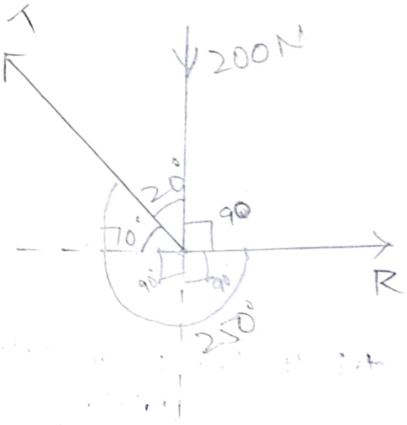
$$\boxed{T_{AB} = 397.91 \text{ N}}$$

(6) A sphere weighing 200 N is tied to a string hanging by a wall as shown in the figure. Determine the tension in the string and reaction from the wall.



Sol





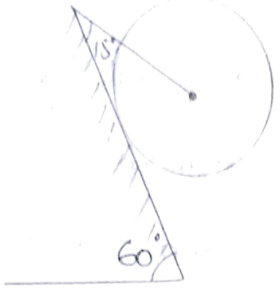
Apply Lami's Theorem

$$\frac{T}{\sin 90} = \frac{200}{\sin 250} = \frac{R}{\sin 20}$$

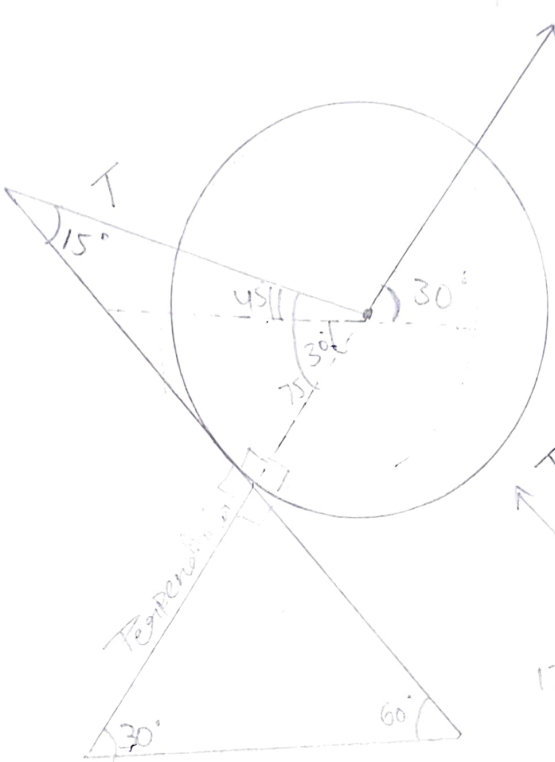
$$\frac{T}{\sin 90} = \frac{200}{\sin 250} \qquad \frac{R}{\sin 20} = \frac{200}{\sin 250}$$

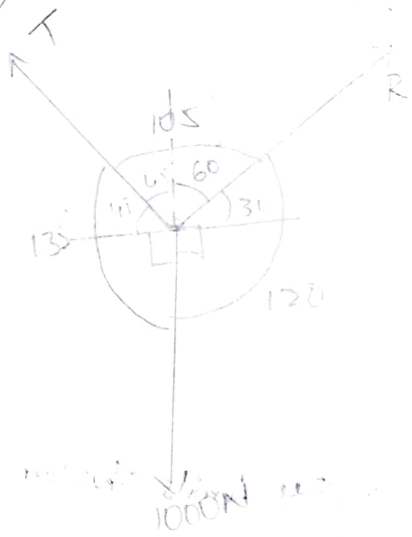
$$\boxed{T = -212.83 \text{ N}} \qquad \boxed{R = -72.79 \text{ N}}$$

① Determine the tension in the string and reaction at the contact surface for a cylinder of weight 1000N Placed as shown in the figure.



Sol





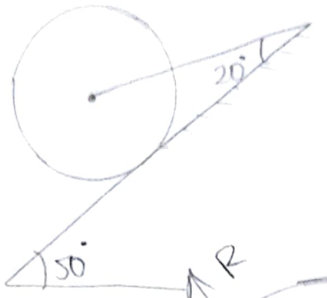
Apply Lami's theorem

$$\frac{T}{\sin 120} = \frac{1000}{\sin 105} = \frac{R}{\sin 135}$$

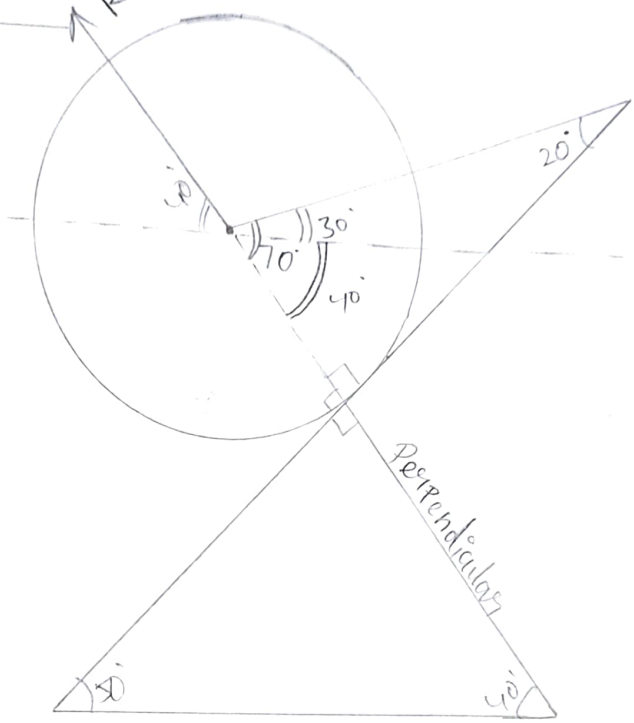
$$\frac{T}{\sin 120} = \frac{1000}{\sin 105} \qquad \frac{R}{\sin 135} = \frac{1000}{\sin 105}$$

$$\boxed{T = 896.5\,N} \qquad \boxed{R = 732.05\,N}$$

(8) Find the tension in the string and reaction at the contact surface for a cylinder of Weight 1000 N placed as shown in the figure using Lami's Theorem.



Sol

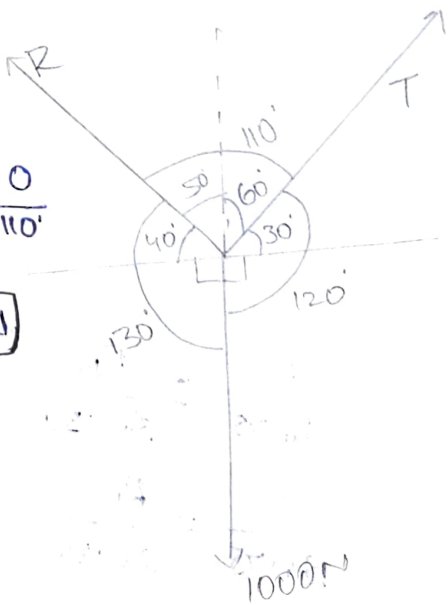


Apply Lami's theorem

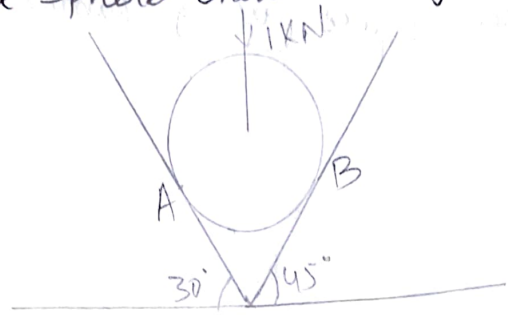
$$\frac{T}{\sin 130^\circ} = \frac{1000}{\sin 110^\circ} = \frac{R}{\sin 120^\circ}$$

$$\frac{T}{\sin 130^\circ} = \frac{1000}{\sin 110^\circ} \quad \bullet \quad \frac{R}{\sin 120^\circ} = \frac{1000}{\sin 110^\circ}$$

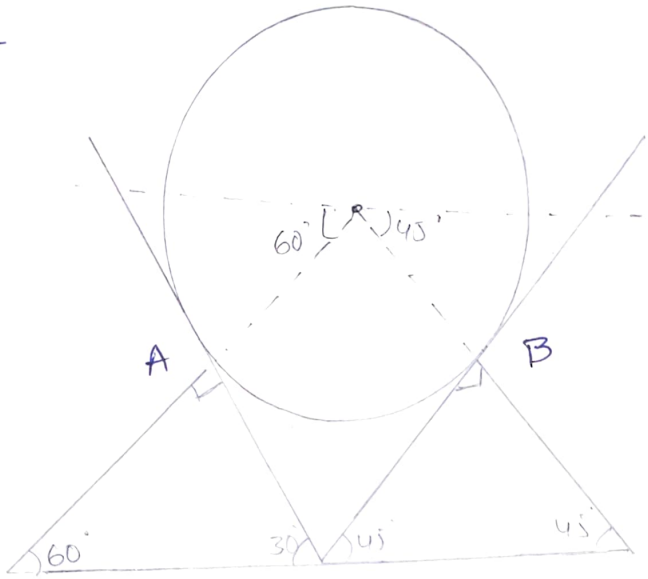
$$\boxed{T = 815.20\,N} \qquad \boxed{R = 921.90\,N}$$

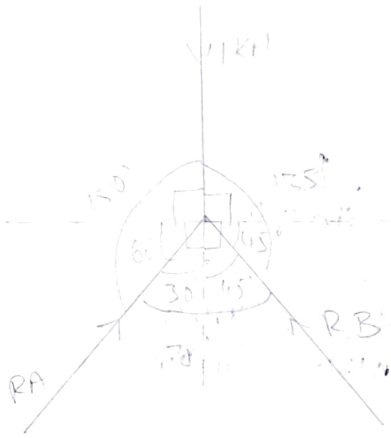


(1) Determine the reactions at the point of contact for the sphere shown in figure.



Sol





$$\frac{1}{\sin 75} = \frac{R_B}{\sin 150} = \frac{R_A}{\sin 135}$$

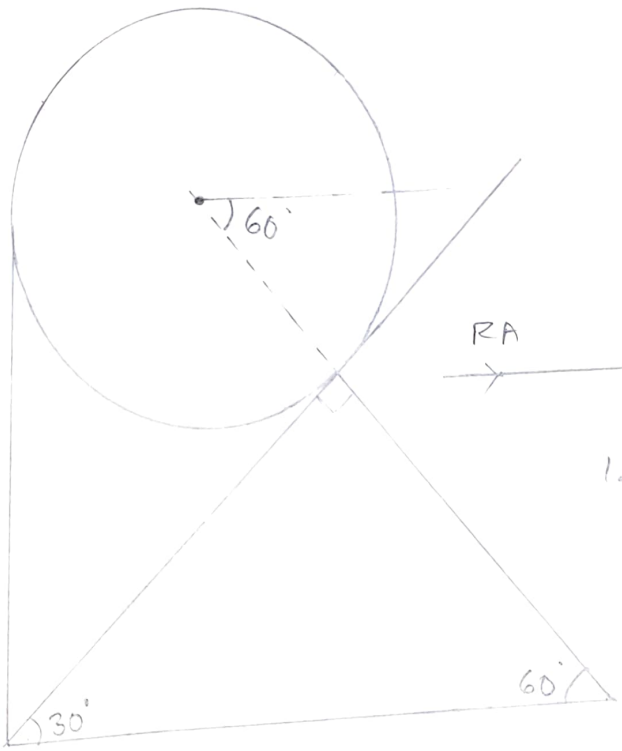
$$\frac{1}{\sin 75} = \frac{R_B}{\sin 150} \qquad \frac{1}{\sin 75} = \frac{R_A}{\sin 135}$$

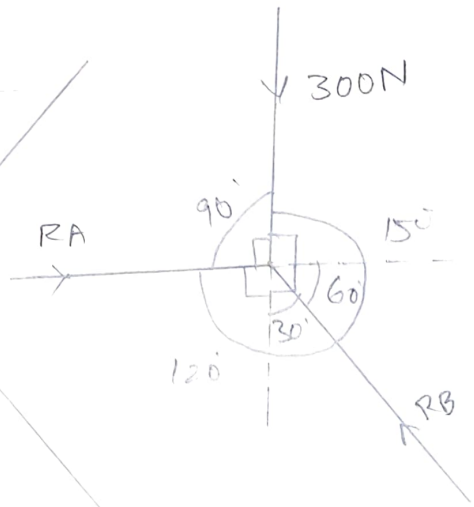
$R_B = 0.511\ KN$ $R_A = 0.732\ KN$

(10) Find the reactions on the sphere as shown in the figure.



Sol



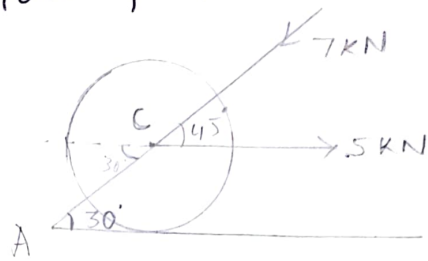


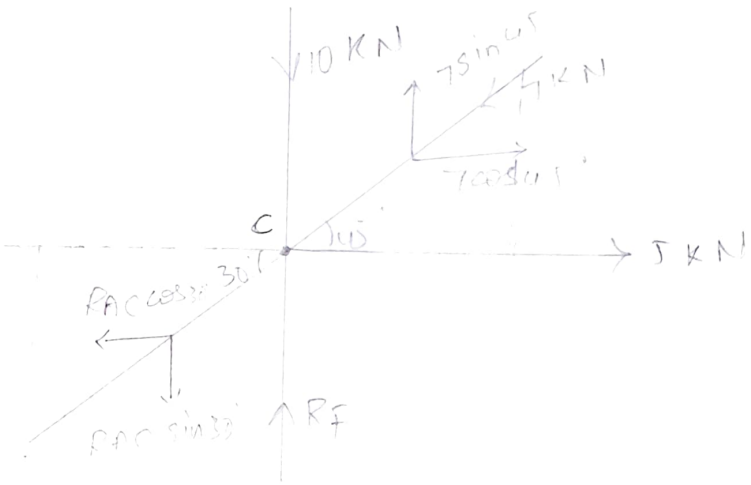
$$\frac{300}{\sin 120^\circ} = \frac{R_B}{\sin 90} = \frac{R_A}{\sin 150}$$

$$\frac{300}{\sin 120} = \frac{R_B}{\sin 90} \qquad \frac{300}{\sin 120^\circ} = \frac{R_A}{\sin 150}$$

$$\boxed{R_B = 346.4\ N} \qquad \boxed{R_A = 173.20\ N}$$

(11) A roller weighing 10 KN rest on a smooth horizontal floor & is connected to the floor by the bar AC as shown in the figure. Determine the force in the bar AC and reaction from the floor, if the roller is subjected to a horizontal force of 5 KN and an inclined force of 7 KN as shown in the figure.





$\Sigma H = 0$

$5 + 7 \cos 45 - R_{AC} \cos 30 = 0$

$$\boxed{R_{AC} = 11.48 \text{ KN}}$$

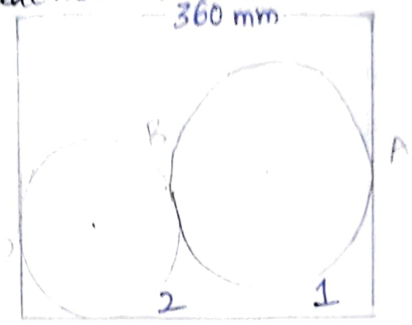
$\Sigma V = 0$

$-10 + 7 \sin 45 - R_{AC} \sin 30 + R_F = 0$

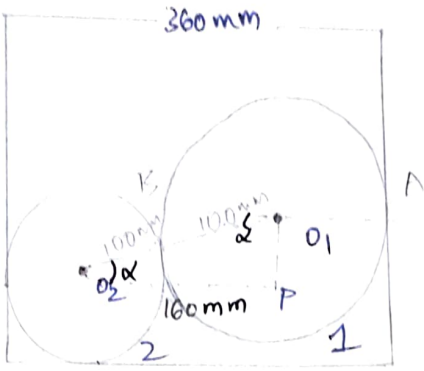
$-10 + 7 \sin 45 - 11.48 \sin 30 + R_F = 0$

$$\boxed{R_F = 10.79 \text{ KN}}$$

(12) Two smooth spheres each of radius 100mm and weight 100N rests in a horizontal channel as shown in the figure. Find the reaction at the contact points A, B, C and D.



Sol



$O_2P = 360 -$ Radius of 1st sphere $-$ Radius of 2nd sphere

$= 360 - 100 - 100$

$O_2P = 160$ mm

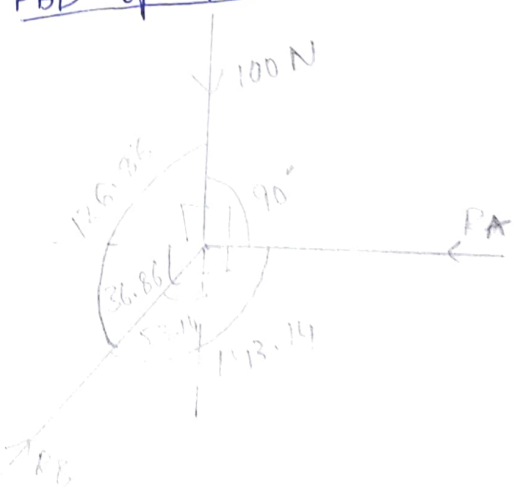
$\cos\alpha = \frac{O_2P}{O_1O_2}$

$\cos\alpha = \frac{160}{200}$

$\alpha = \cos^{-1}\left[\frac{160}{200}\right]$

$\alpha = 36.86°$

FBD of sphere 1



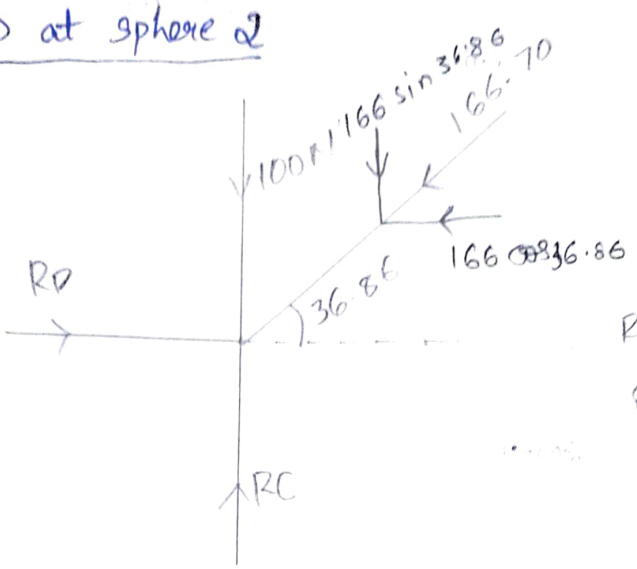
-Applying Lami's Theorem

$\frac{100}{\sin 143.14} = \frac{R_A}{\sin 126.86} = \frac{R_B}{\sin 90}$

$\frac{R_A}{\sin 126.86} = \frac{100}{\sin 143.14}$ $\quad$ $\frac{R_B}{\sin 90} = \frac{100}{\sin 143.14}$

$R_A = 133.38\ N$ $R_B = 166.70\ N$

FBD at sphere Q



$\Sigma V = 0$

$\Sigma H = 0$

$R_C = 199.95\ N$

$R_D = 133.29\ N$

$\Sigma V = 0$

$-100 + R_C - 166 \sin 36.86 = 0$

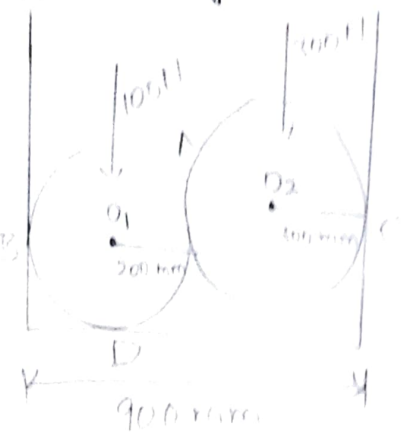
$R_C = 199.57\ N$

$\Sigma H = 0$

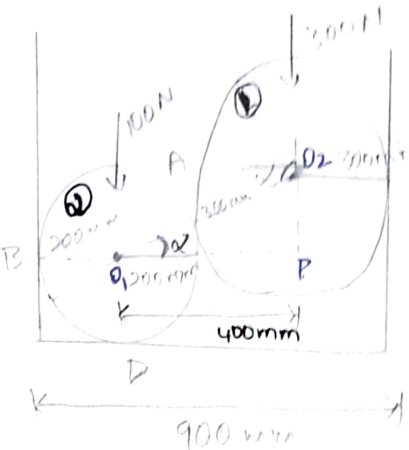
$R_D - 166 \cos 36.86 = 0$

$R_D = 132.81\ N$

(13) Determine the reactions at the contact points as shown in the figure.



Sol



$O_2P = 900 - 200 - 300$

$O_2P = 400mm$

$\cos\alpha = \frac{O_2P}{O_1O_2} = \frac{400}{500}$

$\cos\alpha = \frac{4}{5}$

$\alpha = 36.86$

FBD of sphere 1



Applying Lami's Theorem

$$\frac{300}{\sin 143.14} = \frac{R_C}{\sin 126.86} = \frac{R_A}{\sin 90}$$

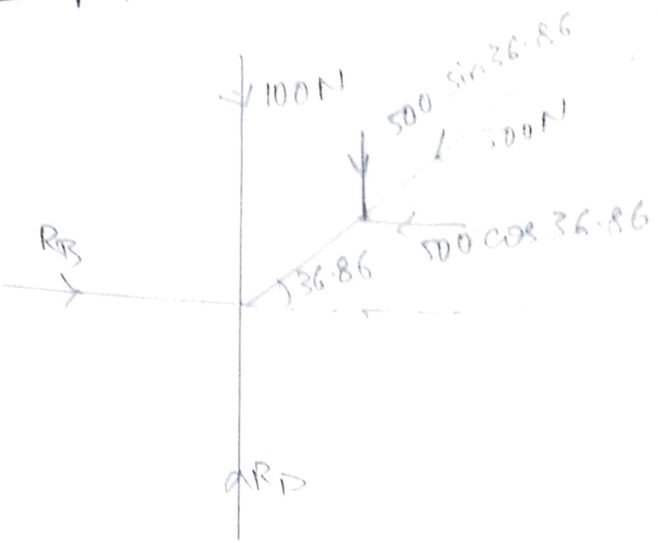
$R_C = \frac{300}{\sin 143.14} \times \sin 126.86$

$R_C = 400.14\,N$

$R_A = \frac{300}{\sin 143.14} \times \sin 90$

$R_A = 500.11\,N$

FBD of Sphere 2



$\Sigma V = 0$

$R_D - 500 \sin 36.86 - 100 = 0$

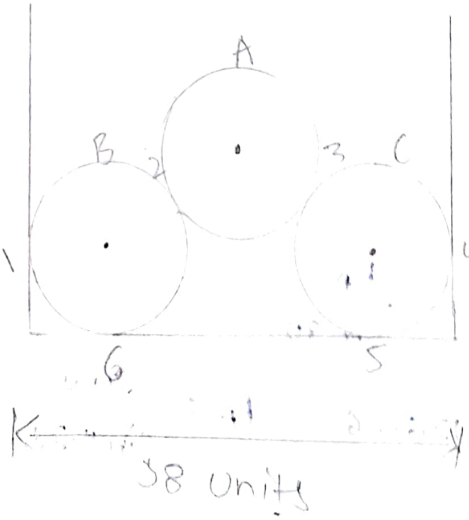
$\boxed{R_D = 400/399.93N}$

$\Sigma H = 0$

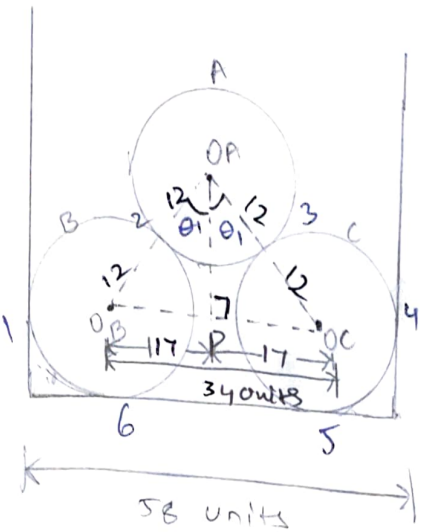
$-500 \cos 36.86 + R_B = 0$

$\boxed{R_B = 400.05\,N}$

(14) Three cylinders weighing 500 N each and 24mm dia are placed in a channel as shown in the figure. Find the reactions at all the contact points.



Sol



$O_B O_C = 58 - 12 - 12$

$O_B O_C = 34$ units

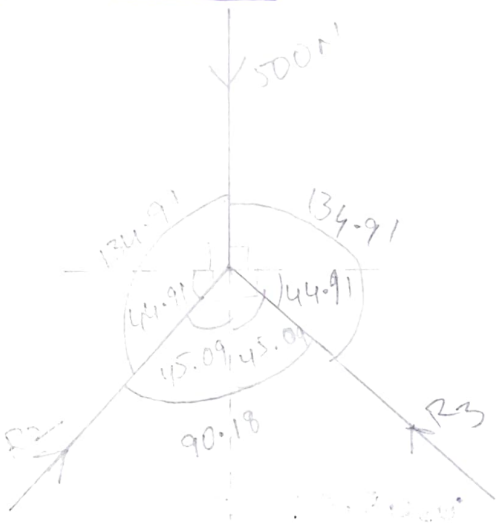
$PO_C = 17$ units

From Δ^{le} $O_A P O_C$

$$\sin\theta_1 = \frac{PO_C}{O_A O_C} = \frac{17}{24}$$

$$\theta_1 = 45.09^\circ$$

FBD at Sphere A

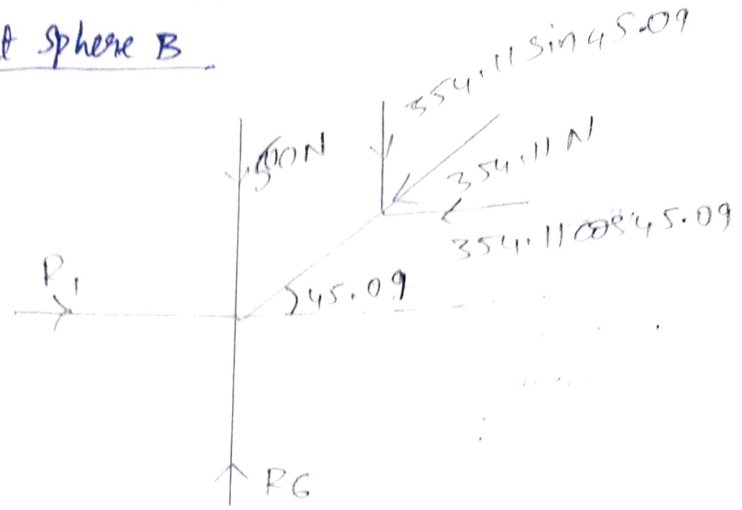


$$\frac{500}{\sin 90.18} = \frac{R_2}{\sin 134.91} = \frac{R_3}{\sin 134.91}$$

$R_2 = 354.11\ N$

$R_3 = 354.11\ N$

FBD at Sphere B



$\Sigma V = 0$

$-500 + R_6 - 354.11 \sin 45.09 = 0$

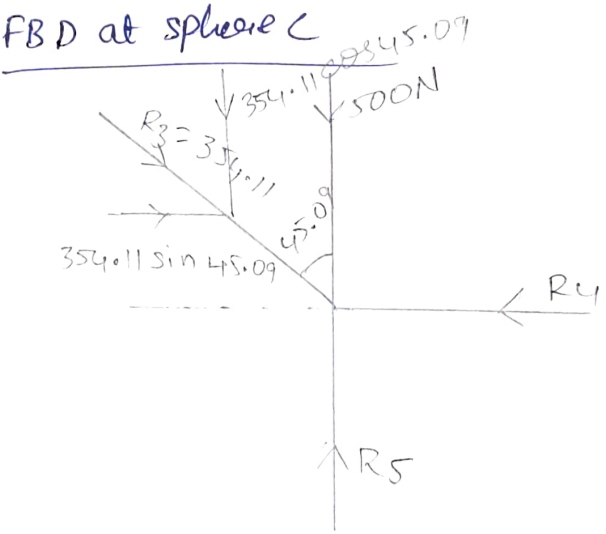
$\boxed{R_6 = 750.78\ N}$

$\Sigma H = 0$

$R_1 - 354.11 \cos 45.09 = 0$

$\boxed{R_1 = 250\ N}$

FBD at sphere C



$\Sigma V = 0$

$-500 + R_5 - 354.11 \cos 45.09 = 0$

$\boxed{R_5 = 750.N}$

$\Sigma H = 0$

$-R_4 + 354.11 \sin 45.09 = 0$

$\boxed{R_4 = 250.78\ N}$